Supplementary Material

Supplementary table 1: Study quality assessment based on Newcastle-Ottawa scale

Author, year	Study type	SELECTION				COMPARABILITY	OUTCOME			SCORE	QUALITY
		Representativeness of the average adult in community	Cohort size	Information on clinical outcomes	Outcome not present at start	factors comparable between the groups	Adequate clinical assessment	Follow up time	Adequacy of follow-up	MAX=8	HIGH>6, MEDIUM 4 to 6, LOW ≤3
Bapaye et al, 2017	Prospective	0	0	1	1	1	1	1	1	6	Medium
Debourdeau et al, 2020	Prospective	0	0	1	1	1	1	1	1	6	Medium
Wong et al, 2020	Retrospective	0	0.5	1	1	1	1	1	1	6.5	High
Kessler et al, 2013	Prospective	0	0	1	1	1	0	1	1	5	Medium
Prasad et al, 2019	Prospective	0	0	1	1	1	0	1	1	5	Medium
Monino et al, 2020	Retrospective	0	0.5	1	1	1	1	1	1	6.5	High
Ortega et al, 2019	Prospective	0	0	0	1	1	1	1	0.5	4.5	Medium

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Patil et al, 2020	Prospective	0	1	1	1	1	1	1	1	7	Medium
Shah et al, 2017	Prospective	0	0	0	1	1	1	0		3	Low
Sumi et al, 2020	Retrospecti ve	0	1	1	1	1	1	1	0.5	6.5	High
Yoo et al, 2020	Prospective	0	0.5	1	1	1	1	1	1	6.5	High

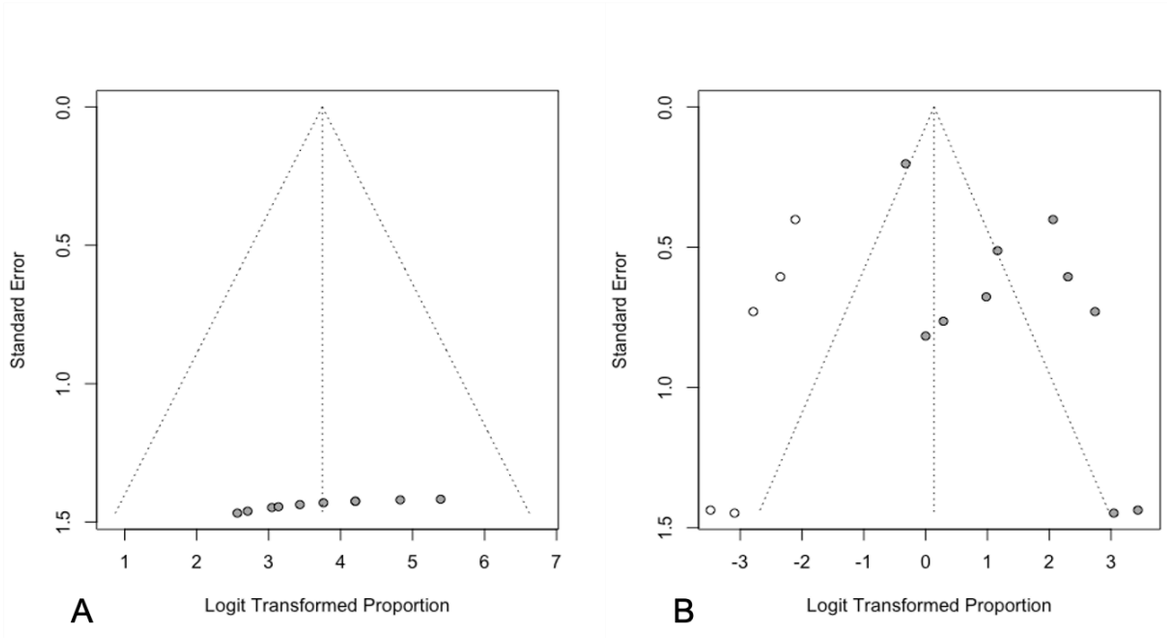


Figure 1: Funnel plot for publication bias

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Appendix 1: Search strategy

Ovid MEDLINE® ALL 1946 to March 08, 2021 &

Embase 1974 to 2021 March 08

1 exp Gastroesophageal Reflux/

2 (GERD or gastroesophageal reflux disease or (refractory adj3 (gastroesophageal reflux or GERD))).mp.

3 1 or 2

4 Endoscopic Mucosal Resection/

5 (antireflux mucosectomy or anti-reflux mucosectomy or antireflux mucosal ablation or anti-reflux mucosal ablation).mp.

6 4 or 5

7 3 and 6

Web of Science (Clarivate Analytics)

Searched three platforms:

Indexes=SCI-EXPANDED Timespan=1965-2021

Indexes=BCI Timespan=1969-2021

Indexes=CPCI-S Timespan=1965-2021

1 TS=("gastroesophageal reflux" OR GERD OR "gastroesophageal reflux disease") OR TS=(refractory NEAR/3 ("gastroesophageal reflux" OR GERD))

2 TS=("Endoscopic mucosal resection") OR TS=("antireflux mucosectomy" OR "anti-reflux mucosectomy" OR "antireflux mucosal ablation" OR "anti-reflux mucosal ablation")

3 1 and 2

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CINAHL Plus with Full Text (EBSCOhost)

S1 (MH "Gastroesophageal Reflux+")

S2 TI (GERD OR "gastroesophageal reflux disease") OR AB (GERD OR "gastroesophageal reflux disease") OR TI (refractory N3 ("gastroesophageal reflux" OR GERD)) OR AB (refractory N3 ("gastroesophageal reflux" OR GERD))

S3 S1 OR S2

S4 TI "endoscopic mucosal resection" OR AB "endoscopic mucosal resection"

S5 TI ("antireflux mucosectomy" OR "anti-reflux mucosectomy" OR "antireflux mucosal ablation" OR "anti-reflux mucosal ablation") OR AB ("antireflux mucosectomy" OR "anti-reflux mucosectomy" OR "antireflux mucosal ablation" OR "anti-reflux mucosal ablation")

S6 S4 OR S5

S7 S3 AND S6

Scopus

1 TITLE-ABS-KEY ("Gastroesophageal Reflux" OR gerd OR "Gastroesophageal Reflux disease") OR TITLE-ABS-KEY (refractory W/3 ("gastroesophageal reflux" OR gerd))

2 TITLE-ABS-KEY ("antireflux mucosectomy" OR "anti-reflux mucosectomy" OR "antireflux mucosal ablation" OR "anti-reflux mucosal ablation") OR TITLE-ABS-KEY ("Endoscopic Mucosal Resection")

3 1 AND 2

Cochrane (Wiley)

Central Register of Controlled Trials

Issue 3 of 12, March 2021

#1 MeSH descriptor: [Gastroesophageal Reflux] explode all trees

#2 (GERD or "gastroesophageal reflux disease" or (refractory NEAR/3 ("gastroesophageal reflux" or GERD))):ti,ab,kw

#3 #1 OR #2

#4 MeSH descriptor: [Endoscopic Mucosal Resection] this term only

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#5 ("antireflux mucosectomy" or "anti-reflux mucosectomy" or "antireflux mucosal ablation" or "anti-reflux mucosal ablation"):ti,ab,kw

#6 #4 OR #5

#7 #3 AND #6

Appendix 2: MOOSE Checklist for Meta-analyses of Observational Studies

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	5
2	Hypothesis statement	-
3	Description of study outcome(s)	7
4	Type of exposure or intervention used	6
5	Type of study designs used	6
6	Study population	6
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	1
8	Search strategy, including time period included in the synthesis and key words	6, Appendix 1
9	Effort to include all available studies, including contact with authors	7
10	Databases and registries searched	6, Appendix 1
11	Search software used, name and version, including special features used (eg, explosion)	Appendix 1
12	Use of hand searching (eg, reference lists of obtained articles)	6
13	List of citations located and those excluded, including justification	9, Figure 1
14	Method of addressing articles published in languages other than English	6
15	Method of handling abstracts and unpublished studies	6
16	Description of any contact with authors	7
Reporting of methods should include		

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17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	6-8
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	6-8
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	6-8
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	9
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	8-10
22	Assessment of heterogeneity	7-8, 12
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	7-8
24	Provision of appropriate tables and graphics	Tables 1-2, Figs 1-4
Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	Figs 2-4
26	Table giving descriptive information for each study included	Table 1 and 2
27	Results of sensitivity testing (eg, subgroup analysis)	11
28	Indication of statistical uncertainty of findings	12-16

From: Stroup DF, Berlin JA, Morton SC et al. for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. JAMA. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008

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Appendix 3: PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3-4
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5-6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	-
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6, Appendix 1
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6, Appendix 1
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Appendix 1

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Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6, Appendix 1
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6-7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	9-10
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7-8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	9-10
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8-9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9, Supp table 1

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Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Fig 2-4
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10-11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	9-10
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	11
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	15
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	1

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: **www.prisma-statement.org**.

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